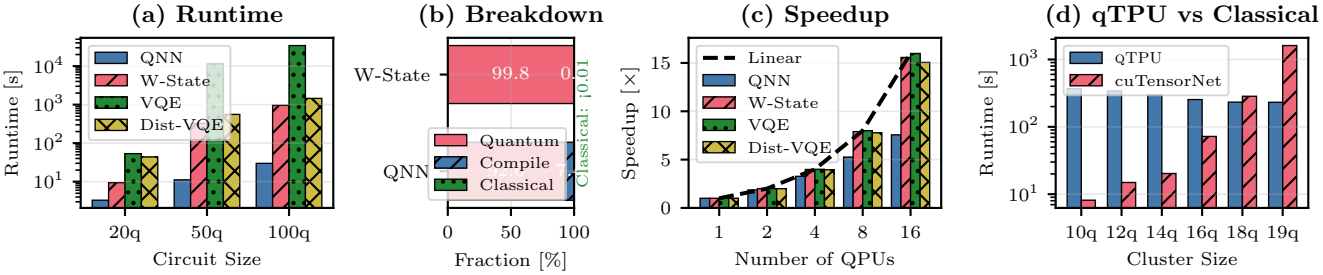
(a) Runtime
Runtime [s]
QNN
W-State
VQE
Dist-VQE
20q
50q
100q
Circuit Size
(b) Breakdown
W-State
QNN
99.8
0.1
Quantum
Compile
Classical
Classical: 0.01
Fraction [%]
(c) Speedup
Speedup [×]
Linear
QNN
W-State
VQE
Dist-VQE
Number of QPUs
(d) qTPU vs Classical
Runtime [s]
qTPU
cuTensorNet
10q
12q
14q
16q
18q
19q
Cluster Size